

# Pathfinder

# From seminar questions to research

## Vincent Danos

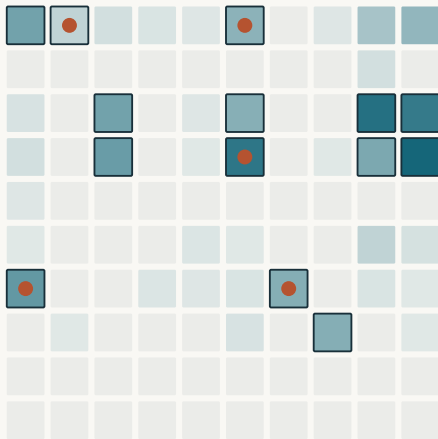
CNRS, ENS

## Elham Kashefi

CNRS, LIP6  
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# William Waites

University of Edinburgh



Ten papers by ten papers.  
Every cell is a possible encounter.

# Convention: mini-abstracts

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## Q mechanism design

### Q4 Incentives to report what one knows

Agents may profit by hiding what they know.  
Rewards can make truthful reporting worthwhile.

### Q7 Sharing resources

Participants compete for limited resources.  
Messages and payments aim to share them well.

### Q3 AI agents in a market

AI agents buy and sell in a simulated market.  
Their performance is compared with humans.

## P systems of cooperating agents

### P6 An outside judge for a team

A team works together; credit is hard to assign.  
An outside AI judges contributions from images.

### P1 Learning about bad outcomes

Learn to act while controlling bad outcomes.  
Estimate losses in the worst fraction of cases.

### P3 A critic for the code reviewer

An AI reviews code; a critic challenges its review.  
The coding agent then revises the program.

# Making connexions ...

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**Bob** “Why use a judge? Could you not ask the agents directly?”

**Alice** “We could ... but wouldn’t they misreport?”

**Bob** “Glad you ask. I have just heard of a mechanism-design paper about that ...”

**Alice** “Interesting!”

## **P6** An outside judge for a team

A team works together; credit is hard to assign.  
An outside AI judges contributions from images.

## **Q4** Incentives to report what one knows

Agents may profit by hiding what they know.  
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# What if we asked across an entire collection?

Mechanism design  $\times$  agent cooperation

|     | P1    | P2   | P3   | P4  | P5  | P6    | P7    | P8   | P9   | P10  |
|-----|-------|------|------|-----|-----|-------|-------|------|------|------|
| Q1  | 1925  | 600* | 300  | 150 | 75  | 1500* | 0     | 75   | 1050 | 1400 |
| Q2  | 0     | 0    | 0    | 0   | 0   | 0     | 0     | 0    | 300  | 0    |
| Q3  | 200   | 0    | 1925 | 0   | 75  | 1575  | 0     | 0    | 3300 | 3000 |
| Q4  | 300   | 0    | 2100 | 0   | 25  | 3150* | 0     | 25   | 1800 | 3575 |
| Q5  | 75    | 0    | 0    | 9   | 0   | 0     | 0     | 0    | 0    | 0    |
| Q6  | 25    | 0    | 0    | 0   | 100 | 50    | 0     | 0    | 600  | 240  |
| Q7  | 2200* | 0    | 0    | 120 | 75  | 150   | 1650* | 0    | 150  | 25   |
| Q8  | 0     | 25   | 0    | 0   | 0   | 200   | 0     | 1650 | 0    | 50   |
| Q9  | 0     | 0    | 0    | 0   | 0   | 0     | 0     | 0    | 0    | 0    |
| Q10 | 0     | 0    | 0    | 0   | 0   | 0     | 0     | 0    | 0    | 0    |



100 possible encounters  
14 investigated pairs  
5 accepted manuscripts

 feasibility  $\times$  gain  
 investigated  
\* internally accepted

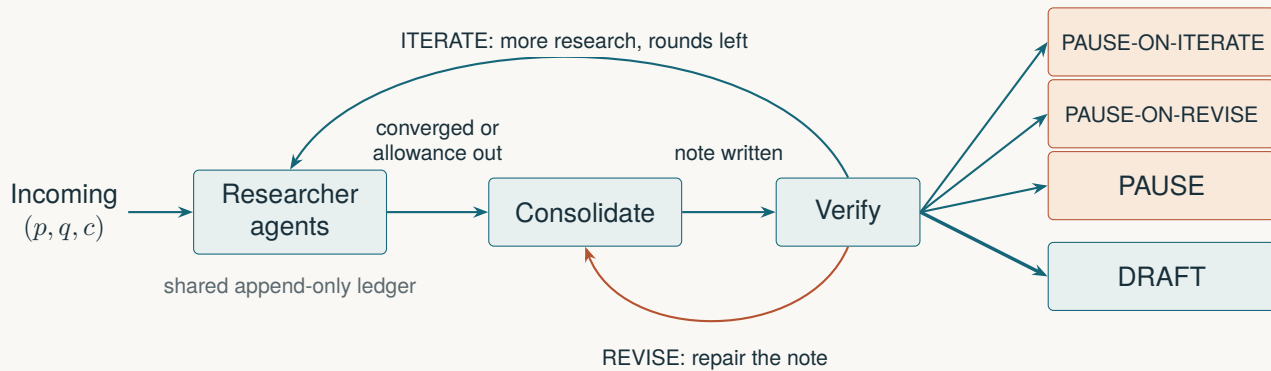
# Scan. Research. Edit.

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The output may be a result, an obstruction, a mistake in either paper, or an account of what is missing to pursue the connexion.

# The research phase



Research question changes after iterations.

Seed  $c$  is a hypothesis to investigate, not a claim the agents must pursue.

Agents are resource-bound.

# Practicalities

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## Nothing is lost

Every step is saved as it happens.

After a crash, work resumes where it stopped.

## Spending is bounded

Every call records its time and its cost.

Each loop has a limit, and the budget is checked before new work starts.

## Reading is reused

Every call starts with the same material.

What the model has already read costs much less to read again.

What an agent reads, in order

|                |                                     |                          |
|----------------|-------------------------------------|--------------------------|
| The two papers | The shared ledger, which only grows | This call's instructions |
|----------------|-------------------------------------|--------------------------|

the same as last time, so cheap to reread

Machinery keeps the work safe and the bill predictable

# Q4P6: when can we dispense with the judge?

## Q4 Incentives to report what one knows

Agents may profit by hiding what they know.  
Rewards can make truthful reporting worthwhile.

## P6 An outside judge for a team

A team works together; credit is hard to assign.  
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### Seed

Replace the judge with the participants' own reports.

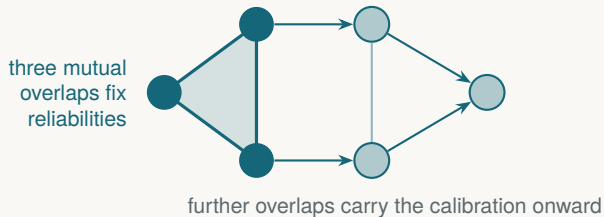
### → Obstacle

Truthful reports can still be insufficient for ranking.

### Conditions

Informative observations. Structured errors.  
Enough shared comparisons.

### → Conditional result



**Under these conditions, peer reports identify the model's ranking without an external judge.**

Population-level, conditional application. A model-defined ranking is not automatically causal contribution.

# Q7P1: a tale of two runs

## Q7 Sharing resources

Participants compete for limited resources.  
Messages and payments aim to share them well.

## P1 Learning about bad outcomes

Learn to act while controlling bad outcomes.  
Estimate losses in the worst fraction of cases.

**Shared seed:** adapt resource sharing to participants who care about bad outcomes.

### First run

Risk approximation  
**Error bound**

Mechanism application assumes  
a suitable mechanism exists

### Rerun

Q7's printed payments  
**Gain 9.5; pay 10.5**

Follow-up: repair the payments  
and strengthen the assumptions

The first run bounded the error in estimating risk, but left the resource-sharing rules conditional. The rerun found that Q7 could charge a participant more than they gained, prompting a repair under stronger assumptions.

**The approximation bound survives. Its intended application needed repair.**

# Q3P3: a promising question, an unfinished test

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## Q3 AI agents in a market

AI agents buy and sell in a simulated market.  
Their performance is compared with humans.

## P3 A critic for the code reviewer

An AI reviews code; a critic challenges its review.  
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### Seed: spot a rush to finish

Q3 finds urgency in traders' language just before a trade. P3's reviewers sometimes agree too soon.

**Could Q3's language analysis warn when P3's reviewers are rushing to agree?**

### Shift: test simpler agents

The researchers found that simple random traders could do well under Q3's rules.

This suggested a different question for P3: **does the critic help, or do extra rounds of review explain the gain?**

### Test: compare fairly

Compare a reviewer working alone with a reviewer helped by a critic.

Let both review and repair repeatedly, on the same tasks, with equal computing budgets.

**The required materials were unavailable.**

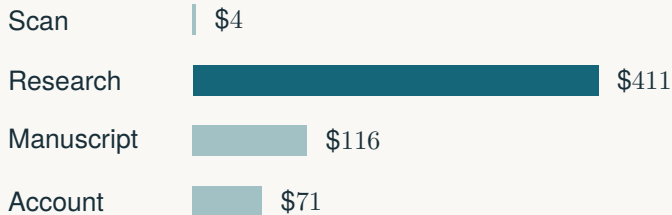
Q3 inspired the test; it did not establish the answer for P3. Neither the seed nor the revised connexion was demonstrated.

Original outcome: PAUSE-ON-ITERATE. Missing evidence, not proof that the critic is ineffective.

# Breadth is cheap. Investigation dominates.

Economics of the 10×10 pilot

**\$601** recorded-token scenario



51.0M recorded input tokens; 1.82M output.  
Flat GPT-6 Standard rates: \$10/M input, \$50/M output.

**43.7 h** campaign span  
Includes pauses and owner continuations.

**8.1 h** summed call time  
Overlapping calls; not wall-clock duration.

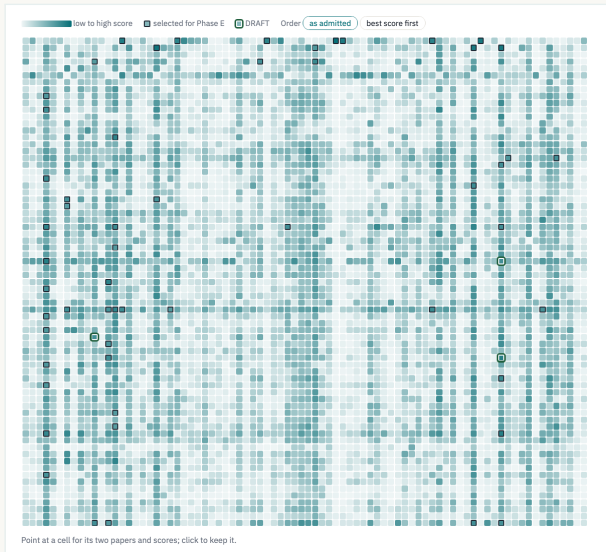
**25 h** illustrative human scan  
Assuming 15 minutes per pair.  
Not a measured human control.

**Scenario, not a bill:** cache usage was incompletely logged; exact repricing is unavailable.

Common tariff, not an observed medium-effort run. Subscription quota is separate. [OpenAI pricing, 16 September 2026](#).

# Scale-up: the Julien Michel validation experiment

Quantum methods for computational chemistry



$71 \times 82$

5,822 scored pairs

59 pairs selected for research

98 finished threads (51 PAUSE, 44 CONTINUE)

3 DRAFT outcomes

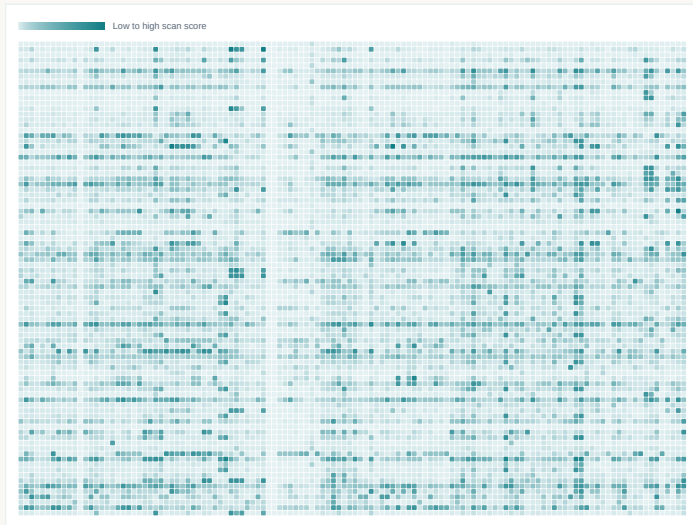
**One pair can suggest several connexions.**

The expert already knows  
one side of the connexion.

Expert assessment becomes a search for directions in their own  
programme.

# Scale-up in quantum: QSL meets vendor applications

124 vendor-side paper artefacts (columns)



# 88 × 124

10,912 candidate pairs

**100** pairs in the original selection

Fully scored by two judges (Opus 5 and gpt-5.6).

Global rank correlation: **0.740**.



Pathfinder: hypothesis generation  
across the quantum  
software-hardware boundary  
Danos, Kashefi, Waites  
Companion manuscript, 2026

## Later E-prime campaign

**147** final notes (144 PAUSE, **3 DRAFT**)

Expanded campaign, not just the original 100 pairs.

Watchboard snapshot: 14 September, 20:42 UTC.

# Questions

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How much of creativity is connexion?

What is the density  $d(Q, P)$  of  $Q \times P$  in good ideas?

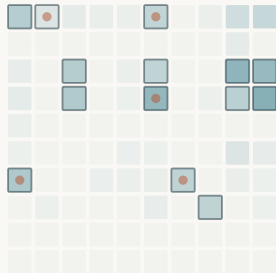
Can agents also help choose which corpora to bring together?

How to trade-off search space restriction and loss rate?

If agents can make the connexions and also investigate what follows ...

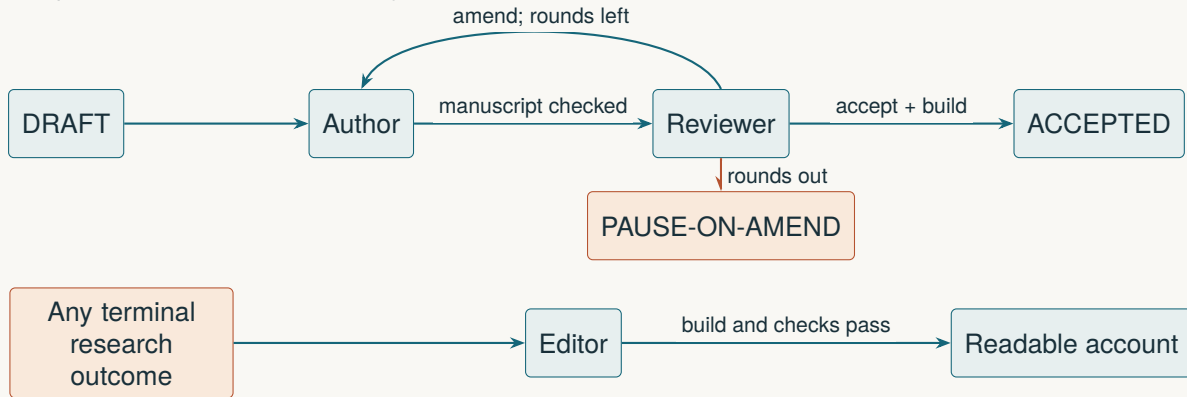
What are the other things humans do in science?

Choosing questions.    Inventing concepts.  
Judging significance.    Applying for funding.



# Backup: editing preserves the research outcome

Two output branches, with different entry conditions



Acceptance refers to the reviewed manuscript version. Neither output branch silently reopens research.

# Backup: what Q4P6 actually assumes

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## **Q4 Incentives to report what one knows**

Agents may profit by hiding what they know.  
Rewards can make truthful reporting worthwhile.

## **P6 An outside judge for a team**

A team works together; credit is hard to assign.  
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- Truthful reporting under Q4's own incentive and belief assumptions.
- Observers are better than chance, with stable symmetric errors.
- Their errors are conditionally independent on the same latent comparison.
- Overlapping observations contain a calibration triangle and connect onward.
- The agent-comparison graph is connected; the ranking model is specified.

A population-identification result, not a finite-sample guarantee or a theorem that these conditions arise automatically.

# Backup: what the dollar scenario does and does not say

Original 330-call cohort; later runs excluded

| Stage                                   | Calls | Summed call time | Scenario USD |
|-----------------------------------------|-------|------------------|--------------|
| Scan                                    | 106   | 18.3 min         | 3.75         |
| Research + consolidation + verification | 168   | 6.07 h           | 410.56       |
| Manuscript preparation + review         | 38    | 1.23 h           | 116.11       |
| Readable accounts                       | 17    | 28.7 min         | 70.52        |
| Probe                                   | 1     | 4.4 s            | 0.13         |

GPT-6 Standard, per million tokens: input \$10; cache read \$1; cache write \$12.50; output \$50.

Only input and output counters were retained in these receipts. Missing cache and per-request context data prevent exact normalisation.

Scenario applies \$10/M and \$50/M to recorded counters only. Above 272K input tokens, the full request has higher rates. Medium effort is not a multiplier that reconstructs a counterfactual run.

# Backup: why did we pause?

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Proposed review distinction, separate from the stopping mechanism

## PAUSE-PLUS

An actionable continuation exists.

Name the missing material or experiment, the next step, and what its outcomes would establish.

## PAUSE-MINUS

Evidence undermines this direction.

Explain what was ruled out and why, without claiming that every possible connection is exhausted.

An exhausted iteration budget is a stopping reason, not a scientific assessment. Allow “undetermined” too.